Apogee Detection

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1 Abstract

*This write-up clarifies various algorithms for apogee detection for the rocket **Zephyrus**, using data from the rocket **Xanthus**. We intend to find the algorithm allowing for fastest, simplest, and most accurate apogee detection.*

2 Proposals

2.1 100 Foot Drop

The latest on-rocket implementation of apogee detection uses a 100-ft drop: barometer data is used to identify altitude, and a 100-ft drop is searched for. This method has been proven to succeed, without record of false positives and missed events. However, it relies on barometer accuracy which could become non-neglibile for higher altitudes.

2.2 Yaw and Pitch Angles

Initially, we propose a measurement of the angle from the vertical,

$$\gamma \equiv \cos^{-1}(\cos(\psi) * \cos(\theta)) \tag{1}$$

where ψ is the yaw, and θ is the pitch (φ is the roll angle). This calculation seems usable:

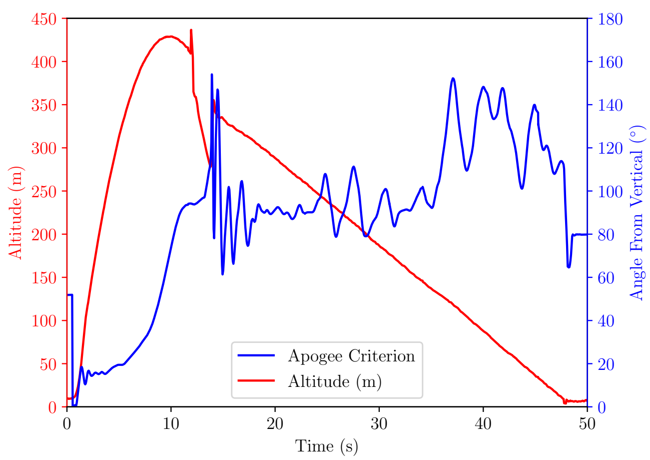

Figure 1. Proposed angle-from-vertical apogee criterion in blue, for 2nd Xanthus launch data.

We can use this to refine the apogee detection, according to the following algorithm:

```
if (altitude > 100m) {
    if (gamma > 45deg) {
        if (smoothGammaDeriv < lastSmoothGammaDeriv) {
            markApogee()
        }
    }
}
```

where the smoothed $\dot{\gamma}$ would be calculated using a 1 s moving average:

```
for i in range(len(gdot)):
    size = int(1/dt) + 1 # half a second
    if i < size:
        gdotsmooth.append(gdot[i])
    else:
        gdotsmooth.append.append(np.mean(gdot[i-size:i]))
```

The performance is shown on the following graph.

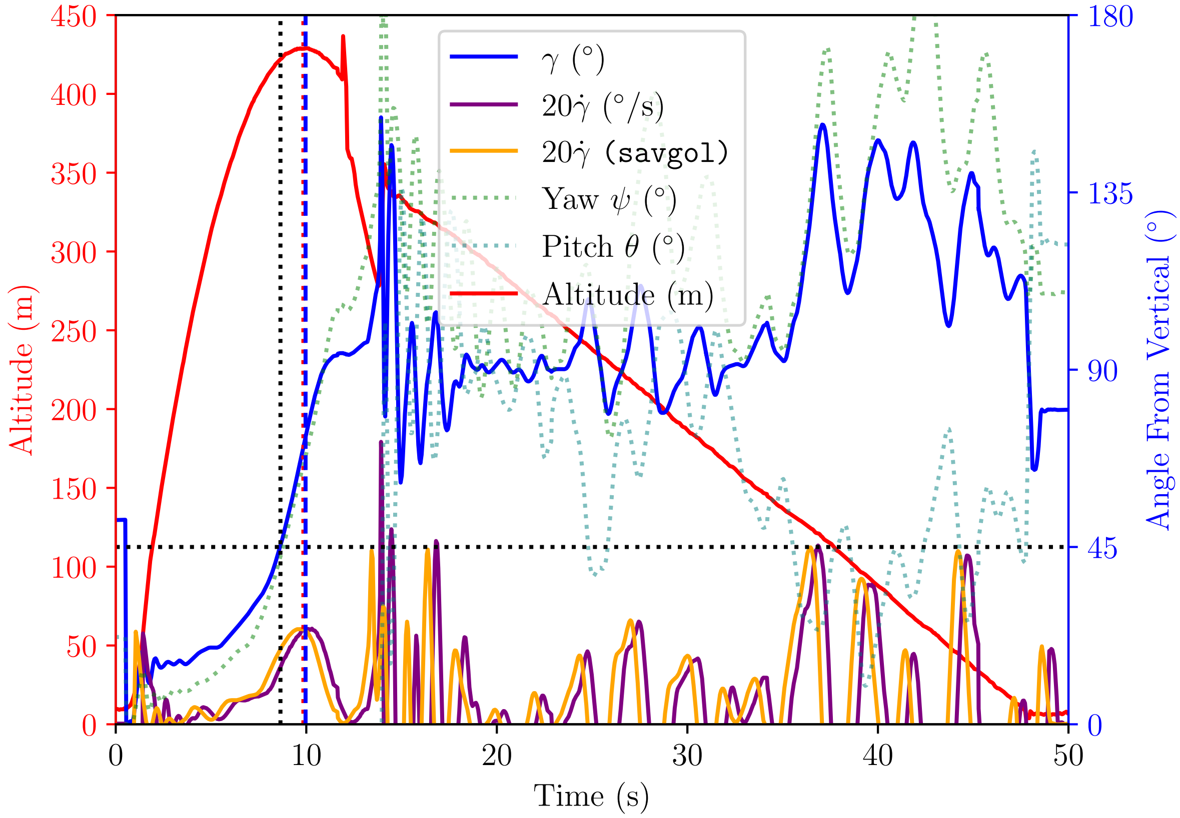

Figure 2. Proposal of Apogee Detection, using the aforementioned angle. The orange line denotes a Savitzky-Golay filtered derivative, essentially a noise-free derivative to highlight accumulated lag from the moving average. In this case, the result is acceptable, and is quite close to the true apogee according to altitude.

We are able to detect apogee $\boxed{2}$ seconds earlier than the 100-ft drop, and we are able to detect apogee to within $\boxed{0.15}$ of the true altitude apogee.